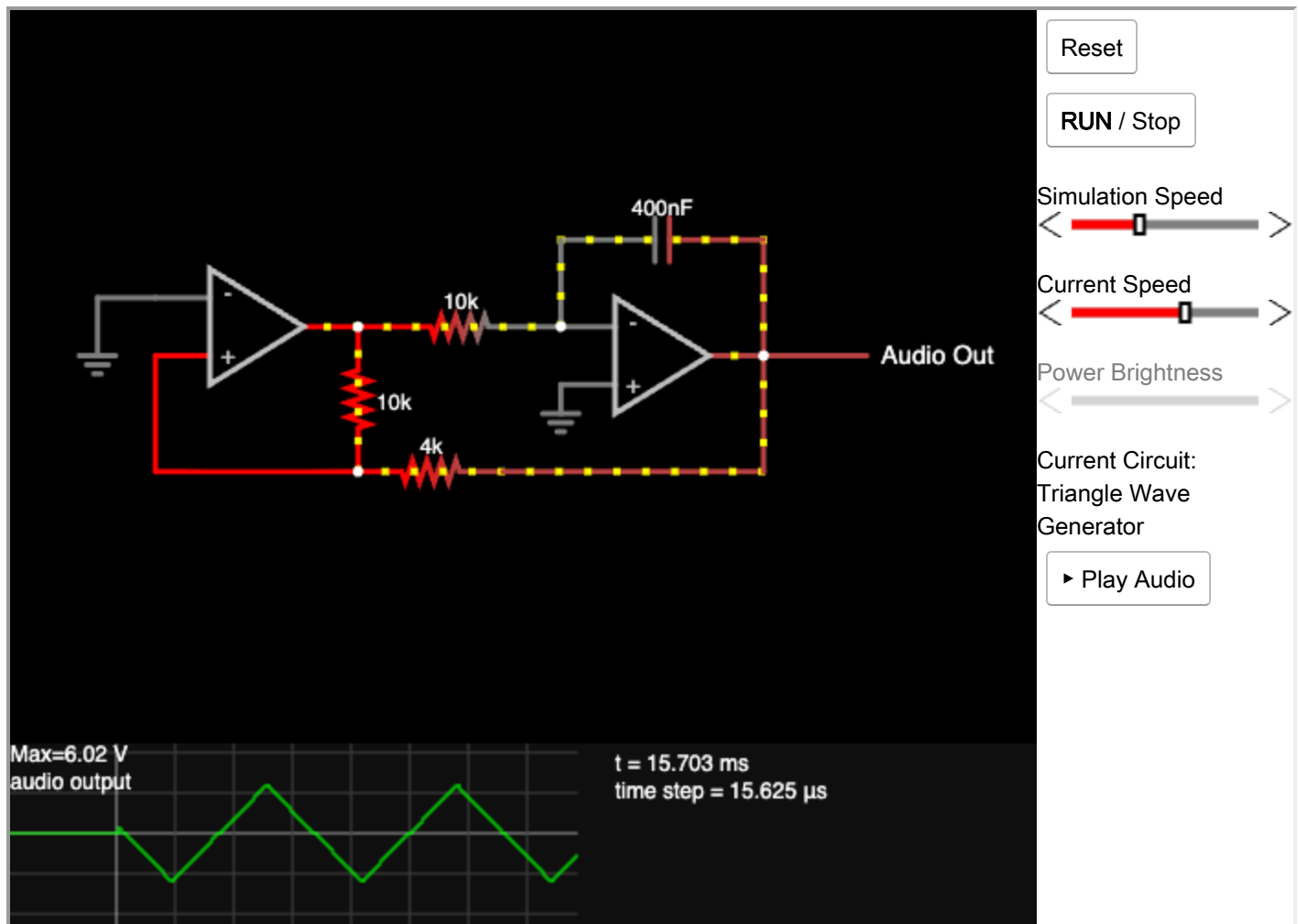




This circuit is an [oscillator](#) that generates a [triangle wave](#).

The second half of the circuit is an inverting [integrator](#). The first [op-amp](#) starts with its two inputs in an unknown state; let's say it starts with + slightly higher than – (which is at ground). The op-amp greatly amplifies this difference, bringing its output to the op-amp's positive power supply voltage, its maximum output (15 V in this case). With this positive input, the integrator's output falls at a constant rate.

The 10k and 4k resistors act as a voltage divider which put the first op-amp's + input 4/14ths of the way from the second op-amp's output to the first op-amp's output. When this input reaches ground, then the first op-amp's output switches polarity, and the integrator switches direction, forming the other half of the triangle. When the first op-amp switches polarity again, a new cycle begins.

To hear the waveform, increase the simulation speed and press the **Play** button.

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